

XIAOCHI XIE

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EDUCATION

Stanford University

M.S. in Mechanical Engineering; GPA: 3.8/4.0

Stanford, CA

Expected Jun 2027

University of Rochester

Bachelor of Engineering in Mechanical Engineering; GPA: 3.9/4.0

Rochester, NY

May 2025

RESEARCH INTERESTS

Bioinspired hydrodynamic and tactile sensing (whisker arrays with fiber-optic and vision-based readout); contact-rich robot design (compliant and stiffness-configurable mechanisms); underwater manipulation and perception in constrained, low-visibility environments.

PUBLICATIONS & MANUSCRIPTS

Published / Accepted

- [1] **Xiaochi Xie**, Pegah Mohammadi, Sobhit Singh. *Pressure-driven structural phase competition and functional response in layered LiInP_2S_6* . *Computational Materials Science*, vol. 275, 115055, 2026. [doi:10.1016/j.commatsci.2026.115055](https://doi.org/10.1016/j.commatsci.2026.115055)
- [2] Hao Li, Long Yin Chung, Jack Goler, Ryan Zhang, **Xiaochi Xie**, Huy Ha, Shuran Song, Mark R. Cutkosky. *UMI-Underwater: Learning Underwater Manipulation without Underwater Teleoperation*. *Robotics: Science and Systems (RSS)*, 2026. [arXiv:2603.27012](https://arxiv.org/abs/2603.27012) · [Project page](#)

Under Review

- [3] **Xiaochi Xie**[†], Hao Li[†], Shixuan Zhao, Siyue Yao, Shuran Song, Mark R. Cutkosky. *Fluorescence-enhanced Whisker Array with Vision-based Deformation Analysis for Underwater Source Localization*. Under review. [†]Equal contribution.
- [4] Hao Li, Tianyu Tu, Siyue Yao, Ziyang Chang, Juhyun Jung, **Xiaochi Xie**, Long Yin Chung, Tian-Ao Ren, Genliang Chen, Mark R. Cutkosky. *Fiber Bragg Grating Whiskers for Bioinspired Hydrodynamic Perception on Underwater Robots*. Under review, *IEEE Transactions on Robotics (T-RO)*. [arXiv:2608.24724](https://arxiv.org/abs/2608.24724), 2026.
- [5] Siyue Yao, **Xiaochi Xie**, Shixuan Zhao, Yutong Li, Hao Li, Mark R. Cutkosky, Genliang Chen. *Lego-Like Stiffness Configuration of Planar Compliant Modules for Task-Specific Flexible Interfaces*. Under review.

RESEARCH EXPERIENCE

Fluorescence-Enhanced Whisker Array for Underwater Source Localization

Stanford, CA

Co-first author; BDML, Advisors: Prof. Mark R. Cutkosky, Prof. Shuran Song

May 2026 – Sep 2026

- Led the development of a five-whisker fluorescence-enhanced sensing array—fabrication, sensing platform, experiments, and data pipeline—that localizes underwater hydrodynamic sources from monocular-camera observations of whisker deformation under low-power (~ 100 mW) UV illumination.
- Developed a two-stage molding process for compliant whiskers with superelastic nitinol cores and fluorescent urethane shells (seal-inspired wavy, tapered, elliptical profile) and built the compact camera/UV sensing platform.
- Designed and ran gantry-thruster pool experiments over a $600 \text{ mm} \times \pm 30^\circ$ region (124 test positions; 8,640 training samples); worked across the deformation pipeline and lightweight CNN (algorithm optimization led by the co-first author): 88 mm mean localization error, with real-time moving-source tracking.

FBG Whiskers for Bioinspired Hydrodynamic Perception

Stanford, CA

Graduate Researcher / Co-author (T-RO study); BDML, Advisor: Prof. Mark R. Cutkosky

Nov 2025 – Present

- Developed Python pipelines for data collection, synchronization, and signal processing across gantry, foil-wake, ROV, and field experiments; analyzed FBG flow responses ($0.1\text{--}0.6$ m/s, angle of attack, wake signatures); supported online branch selection (17/20 correct ROV decisions); contributed analysis, figures, and writing.

Lego-Like Stiffness Configuration of Planar Compliant Modules

Stanford, CA

Co-author; BDML, Advisor: Prof. Mark R. Cutkosky

Jun 2026 – Sep 2026

- Owned fabrication and experiments for a stackable compliant-module framework (three geometries, unified stiffness model, GA + SQP inverse design); 3D-printed and assembled the module stacks.
- Built an optical-breadboard stiffness-characterization platform (linear stage, load cell, ball-joint fixtures); measured all six principal stiffness components ($R^2 \geq 0.99$), matching simulation within 6.5%.
- Implemented the flexible-wrist demonstrations on a robot arm, showing task-dependent contact responses and obstacle interaction at 1 m/s with $\sim 15^\circ$ angular compliance.

UMI-Underwater: Learning Underwater Manipulation

Stanford, CA

Co-author (RSS 2026); joint project, Profs. Mark R. Cutkosky & Shuran Song

Jan 2026 – Present

- Collected the majority of an 800-demonstration on-land dataset (6 object types, 8 backgrounds) for depth-based affordance learning; created the manuscript figures.
- Contributed to self-supervised underwater data collection, land-to-water affordance transfer, and diffusion-policy learning.

Pressure-Driven Phase Competition in Layered LiInP_2S_6 (DFT)

Rochester, NY

Research Assistant / First author, Advisor: Prof. Sobhit Singh

May 2025 – Oct 2025

- First-author study (published in *Computational Materials Science*): performed DFT calculations (VASP; PBEsol + D3, HSE06) of pressure-dependent structural, mechanical, electronic, and optical properties of three polymorphs and wrote the first draft; completed revisions with co-authors through publication.
- Identified a $C2/c$ to $P\bar{3}1c$ in-layer transition at 0.38 GPa; band gaps and absorption edges within each phase change by $\leq 8\%$ up to 10 GPa; computed elastic constants, stability, wave velocities, and Debye temperature, and simulated XRD/Raman/IR fingerprints.

ADDITIONAL EXPERIENCE

Chery Automobile Co., Ltd., Vehicle Technology Department

Wuhu, China

Mechanical Engineering Intern: vehicle-component FEA for hood and door structures

May 2024 – Aug 2024

University of Rochester

Rochester, NY

Teaching Assistant, ME 280: Introduction to Materials Science

Aug 2024 – Dec 2024

TECHNICAL SKILLS

Programming:	Python, MATLAB, OpenCV, PyTorch; data acquisition and signal processing
Experiments:	FBG sensing (multi-channel acquisition), force and stiffness calibration, ROV and gantry experiments, PIV setup, AprilTag tracking
Design / Fabrication:	CAD (SolidWorks, Onshape, Siemens NX), silicone/urethane molding, 3D printing, CNC machining, compliant mechanisms
Simulation:	CFD (ANSYS, Siemens NX), FEA (Nastran), MuJoCo, Isaac Lab, OpenSai
Scientific Computing:	VASP, MechElastic, Phonopy, PyProcar
Languages:	Mandarin Chinese (native), English (fluent)